

Creating Your Personal Race Strategies

Creating your personal race strategies is the final piece of the training process leading up to an event. Your personal race strategies bring your hard-earned training to life on race day. Setting race goals, deciding on nutritional strategies, and choosing your crew and pacers are the last pieces you complete in the weeks before the event. Many times, I have my athletes address the finishing touches during their taper, as a means of providing an outlet for their pent-up energy. In any case, last is certainly not least! By being clear on your goals, nutrition, and pacing, and by choosing the right crew (or not having one at all), you maximize the impact of the work you have done and the fitness you have attained.

Step 0: Review Your Training

Your personal race-day strategies start with a little homework: go review your training. Regardless of whether you keep a paper training log, work up your own spreadsheet, or use technology such as Strava, TrainingPeaks, Movescount, or Garmin Connect, pore over the last several months of training. Find where you were strong, and find the places where there were chinks in the armor. Look at the paces, grades, food, and fluids you utilized in training. This homework serves as the basis from which you will derive your race-day strategies. The effectiveness of your training sets a realistic framework for your goals. The nutrition you used in training is the platform you use to build your race-day nutrition strategy. The paces and workouts you accomplished in training provide a guide for meting out your race-day effort. Examine what you have done, and it will give you some boundaries for what you are capable of and what you need to do come race day.

What Is Your Race-Day Goal?

“What does success look like to you?” is the question I ask my athletes as the season is taking shape. The answer to this question helps me understand an athlete’s goals for the season and for particular events. While I have some influence on shaping those goals, they ultimately belong to the athlete. After all, it’s the athlete who has to put one foot in front of the other come race day. They should run, train, and race to accomplish goals they have created and own.

When narrowing down your race-day goals, I recommend using a similar philosophy. The goals you have on race day are not your partner’s, your children’s, your boss’s, or your training buddy’s. Your family, friends, and peers are key parts of your support network, but in the end your goals are yours and no one else’s. This mindset may seem selfish, but starting from this reference point allows you to focus on you first.

Next, proceed to the practical steps of race-day goal setting. Whereas broad, early season goal-setting questions about what success looks like are purposefully subjective and may seem hippie-dippie, the goals you establish for race day are concrete, actionable, and focused on the tasks at hand. This does not mean they need to be cold and formulaic. “Run with a smile” is just as concrete, actionable, and task-focused as “eat 200 calories an hour.” Both of these are valid goals, and being creative with your goals helps personalize them. They do not have to fit into the stereotypical construct of a particular time, pace, or placing. They do, however, need to describe two things:

- **YOUR OUTCOME GOAL:** This is what you ultimately want to achieve.
- **YOUR PROCESS GOALS:** These are the things you need to do during the race to achieve your outcome goal.

OUTCOME VS PROCESS GOALS

Outcome goal: What you ultimately want to achieve.

Process goals: What you do along the way to support the outcome goal.

Example outcome goal: Finish the American River 50 in under eight hours.

Process goals:

- Eat a smart breakfast of 600 calories.
- Begin the race running 8:00 to 8:30 min/mi.
- Eat early and often, around 250 calories per hour.
- Stay positive.
- Tell the crew “thanks.”

Example outcome goal: Finish the Vermont 100 having enjoyed the day.

Process goals:

- Keep the pace easy and manageable.
- Tell the crew to keep you smiling.
- Thank the volunteers at the aid station.
- Encourage other runners.
- Focus on gratitude that you get to run.

OUTCOME GOALS

I am a big fan of all sports. Although I have a particular affection for my hometown Dallas Cowboys, Dallas Mavericks, Texas Rangers, Texas A&M Aggies, and Dallas Stars, my love of sports and competition is expansive. My biggest fascination with sports is the simple outcome at the end of any sporting event. While I enjoy seeing how a particular play in football materializes and how all the plays throughout the game pile up to create a storyline by the fourth quarter, the outcome of the game is what fascinates me most.

Sports are inevitably an outcome-oriented affair. Wins and losses pile up at the top of the box score; the statistics are underneath. Particularly in running, where there is a starting line, a finish line, and a clock, there is always an outcome for every runner in the race. We can decide to place more or less importance on the outcome vs the process, but the outcome goal is the starting point for goal setting. All properly set outcome goals share three common characteristics that serve as a framework for formulating your particular goals:

- Outcome goals describe the outcome.
- Outcome goals are achievable.
- Outcome goals are challenging.

OUTCOME GOALS MUST DESCRIBE THE OUTCOME. Words matter, and so does specificity. The most important property of an outcome goal is that it must accurately describe the desired outcome. Outcomes can be times, places, or actions. They can be specific (such as a time or placing) or broad (such as a time range). Examples include:

- Place in the top third of the field.
- Finish the race.
- Run under nine hours for a 50-miler.

These are acceptable outcome goals that accurately describe some phenomenon that will occur at the conclusion of the event. Sounds simple? Apparently not. I consistently find athletes and coaches who use the wrong language to describe outcome goals. “I just want to finish” is a common way of expressing that you want to complete the race. I would never let an athlete express their goal this way, and you shouldn’t do it either. The word “just” demeans the process, and “I want” is not an outcome of the event. “Finish the race” is the right goal; it describes the outcome you want to achieve.

RISKY BUSINESS: BALANCING CHALLENGING GOALS AND AFFINITY FOR RISK. Successful outcome goals strike a balance between being achievable and offering a challenge. Where you sit on the “achievability teeter-totter” depends on your individual tolerance for risk. As you set goals that are more challenging and closer to the limits of your capabilities, you must simultaneously accept a higher level of risk associated with those goals. However, goals beyond your capabilities are not well-constructed goals.

It is also important to realize that if you have a low tolerance for risk, an extremely challenging goal is just as inappropriate as a goal that is way beyond your capabilities. In other words, if you are risk averse, your goals will need to be more within your comfort zone. Being risk-averse is not a character flaw; nonetheless, your acceptance of or aversion to risk affects the way you need to set goals. A risk-averse athlete must put a premium on ensuring the goal is within his capabilities, even on a day when he underperforms for whatever reason.

Figure 16.1 illustrates the ideal balance. You have selected a goal that is challenging, and your affinity for risk is high enough that you will make the decisions and take the chances necessary to achieve your goal.

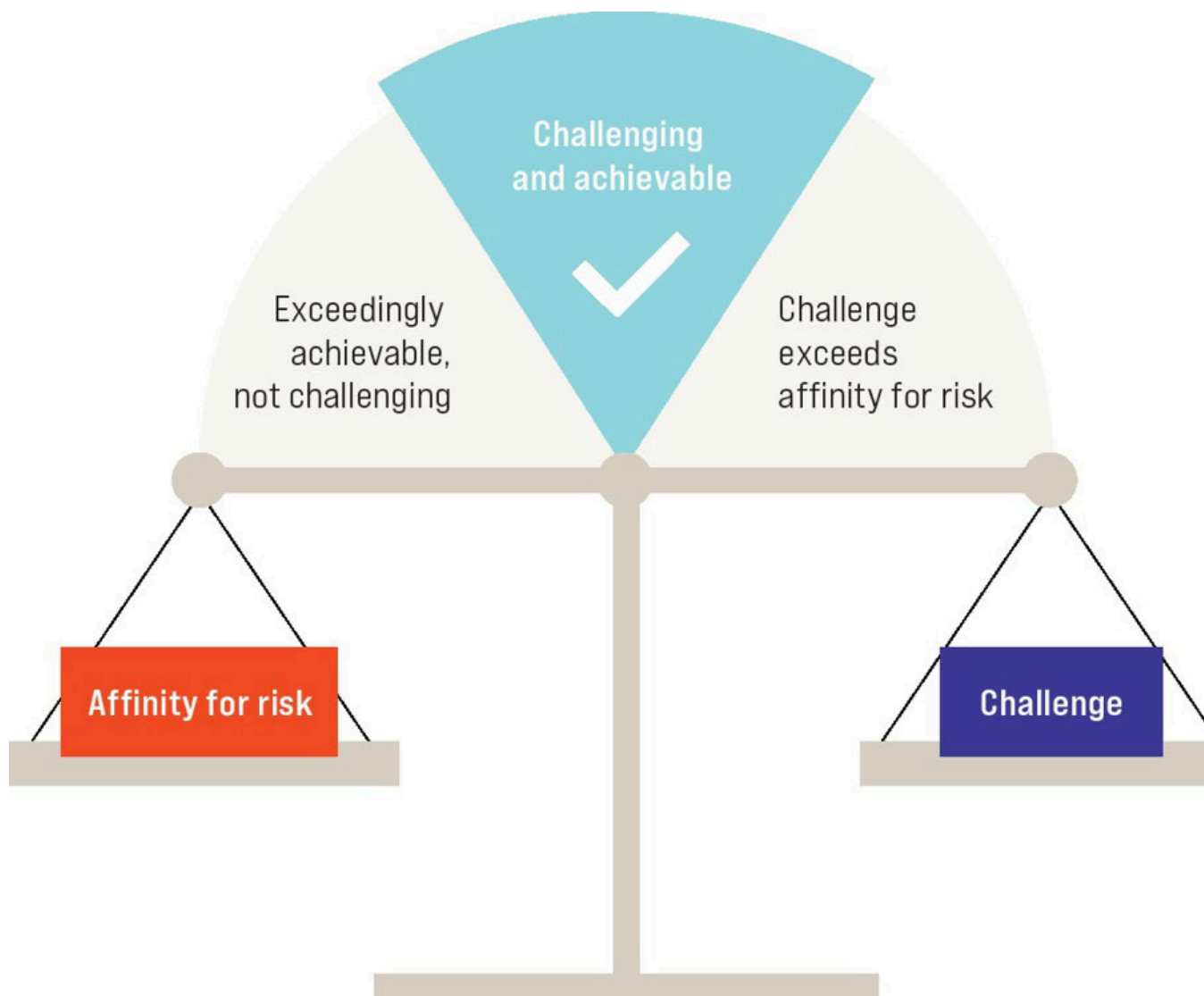


FIGURE 16.1 A goal where the challenge is balanced with the affinity for risk.

A goal that is more challenging requires more risk to remain in balance, as illustrated in Figure 16.2. Put another way, an athlete with greater affinity for risk can pursue goals that present greater challenges.

A goal that is less challenging remains highly achievable for athletes with less tolerance for risk, as shown in Figure 16.3. This can be a good scenario for a beginner because affinity for risk in endurance sports typically increases with experience, which consequently helps make more challenging events increasingly achievable over time.

Figure 16.4 illustrates a scenario in which your goal is highly challenging but you're not tolerant enough of the risks necessary to make that goal achievable. In this situation it is highly unlikely you will achieve your outcome goal. While I encourage athletes to take on challenging goals, you can only push the challenge aspect of your goals as far as your risk tolerance will let you. You may be physically capable of achieving a more challenging goal, but you will fail over and over again unless your affinity for risk is high enough to enable you to fully apply that physical capacity to the goal.

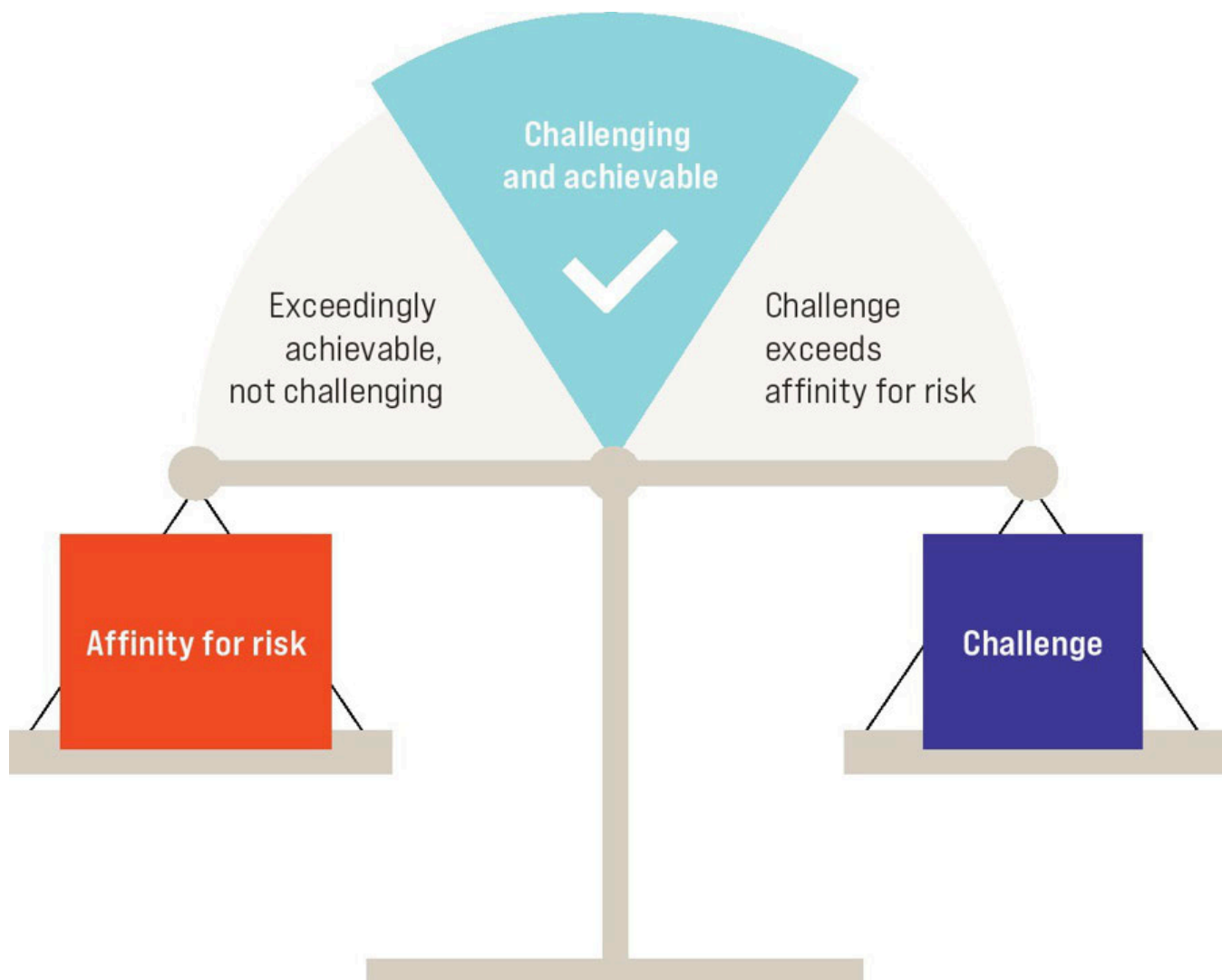


FIGURE 16.2 A bigger challenge that is balanced with a larger affinity for risk.

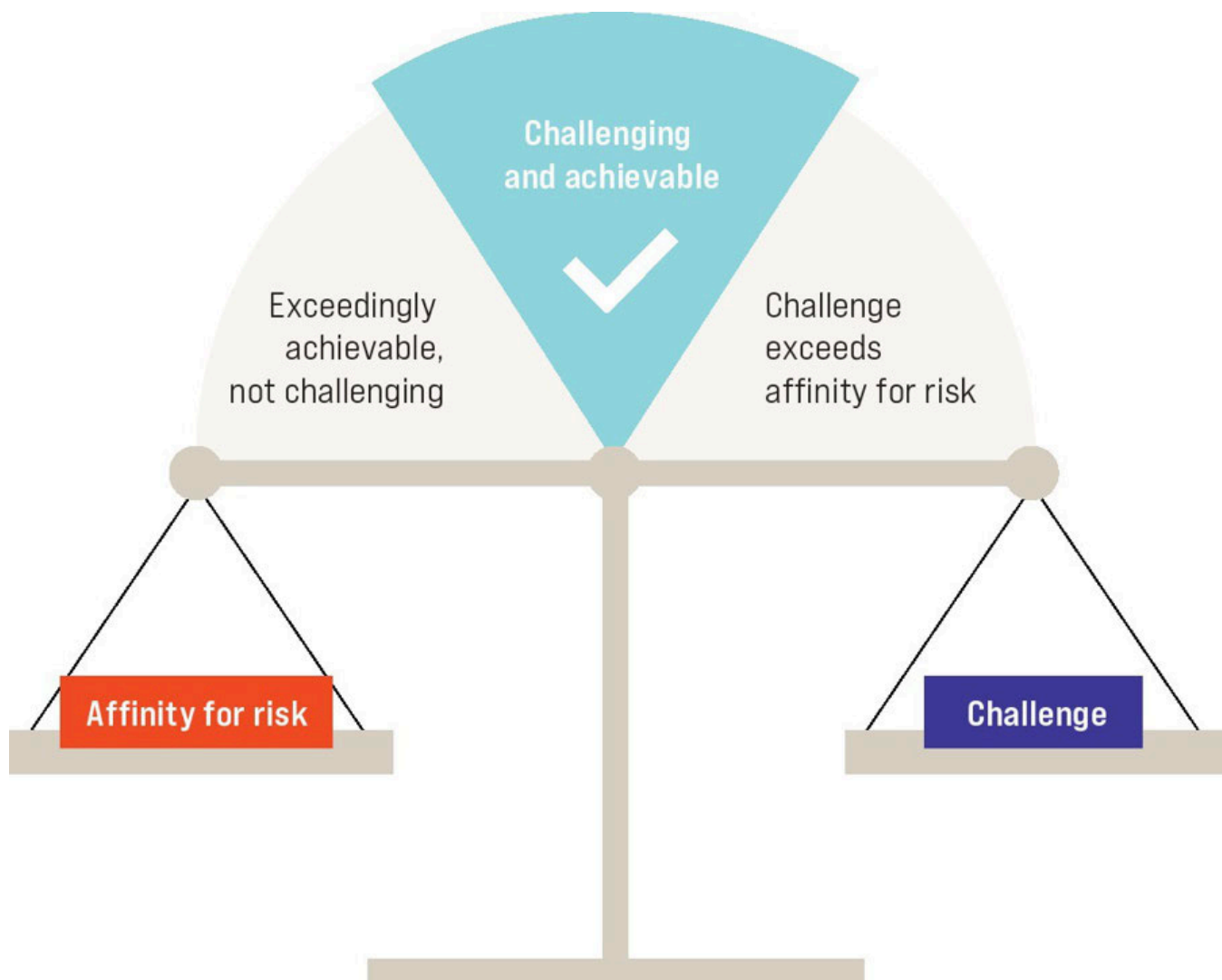


FIGURE 16.3 An easier challenge that is balanced with a smaller affinity for risk.

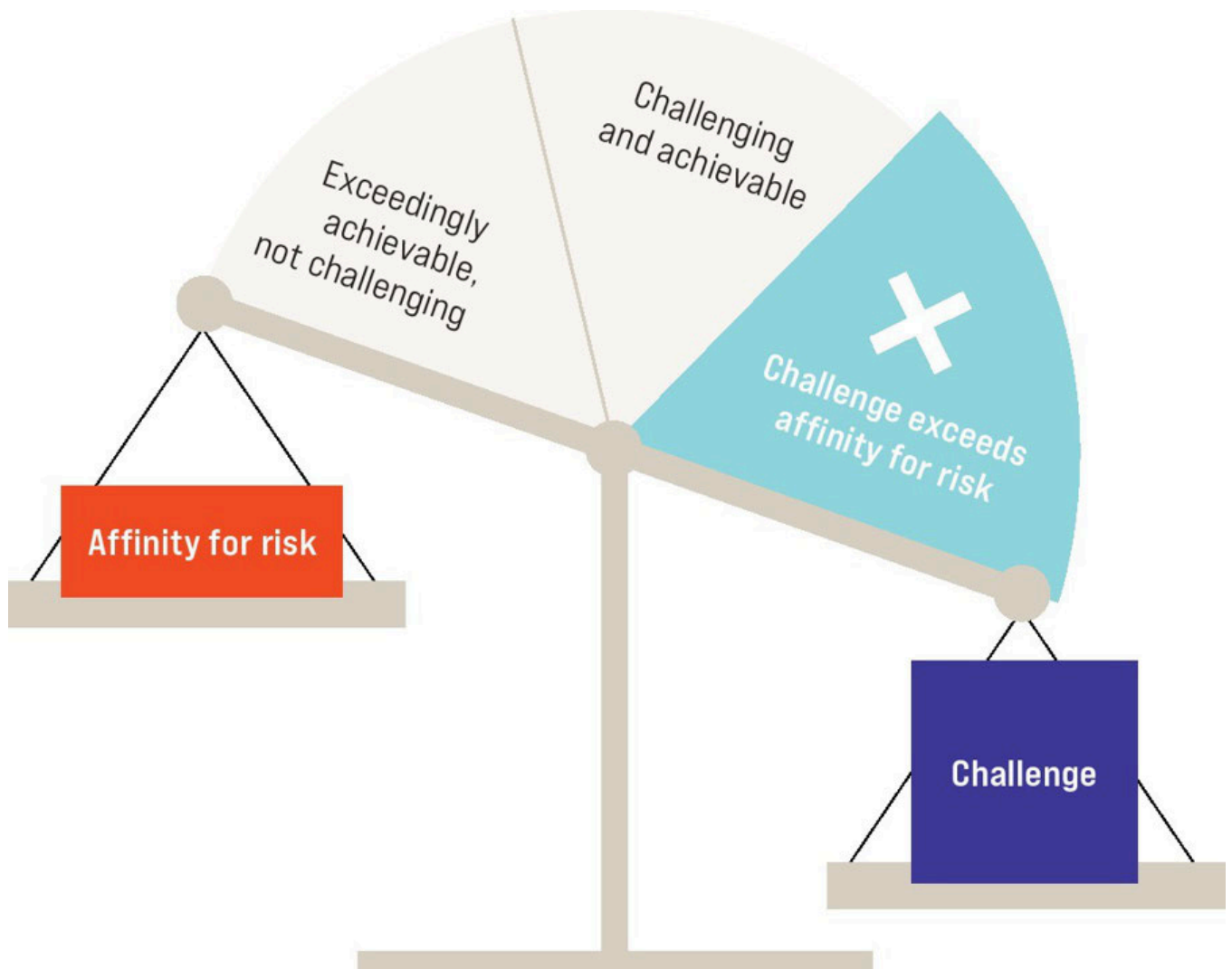


FIGURE 16.4 A bigger challenge that is not balanced with a larger affinity for risk.

Finally, Figure 16.5 illustrates a scenario in which you have a high affinity for risk and your goal is highly achievable. It is virtually guaranteed that you will achieve your outcome goal. Although that may seem like a winning scenario, from a goal-setting perspective it is out of balance. The goal is simply not challenging enough.



FIGURE 16.5 An easy challenge that is thrown out of balance with a large affinity for risk.

What's wrong with an easy goal? Beyond the fact that you're disrespecting your own abilities, goals that are too easy lead to complacency, lack of focus, and big mistakes. Challenging goals force you to focus because many aspects of training, nutrition, and strategy have to go right in order for you to succeed. When the goal is too easy, you don't take it seriously, you don't prepare, and you can find yourself in a surprisingly dangerous position on race day.

HOW MANY OUTCOME GOALS CAN YOU HAVE? When setting outcome goals for an event, think simple and few. A large number of complex outcome goals is a recipe for chaos. I push, prod, and facilitate the narrowing process with my athletes, but in the end, it is their responsibility to craft their own outcome goals. Typically, after a few conversations, one singular outcome-based goal emerges. Any previous iterations of the outcome-based goal normally transition to process goals.

When an outcome goal is a placing or a time in the particular event, most athletes will choose to set A, B, and C goals for that time/place. For example, if your outcome goal is to run nine hours in the American River 50, you might have an A goal of 8:45, a B goal of 9:00, and a C goal of 9:15. These are all still challenging enough and within a normal range of performance. Other athletes will have an A goal of 9:00 and a B goal of finishing the race. Finally, some athletes will have only one goal: "I'm going to run 8:45 or bust!" When this is the case, my role as a coach is to first ensure that the goal is reasonable and balanced with the right affinity for risk. Then I offer 100 percent support for the athlete in that goal.

Once again, these are your goals. Take the time and care to craft them, and you will be on the right path.

Process Goals

I fully admit that I am a mixed martial arts (MMA) fanatic. A sport that started with obscure “last man standing” types of tournaments has now made its way into the mainstream, and I simply cannot get enough of it. I appreciate the sport for the entertainment; it’s a good product and consistently fun to watch. But what really piques my interest is the way these athletes from varying backgrounds, much like ultrarunners, prepare to compete. Training for MMA involves mastery of many different fighting forms. Boxers, wrestlers, Muay Thai fighters, kickboxers, and jujitsu practitioners come together and do battle in the octagon. In these matches, victory can be won in a variety of ways. You can win by forcing your opponent to “tap out,” which is the grown-up, professional version of crying uncle. You can win by decision, which puts the outcome in the hands of judges sitting ringside. You can win by knockout or a referee stoppage.

As these athletes prepare for competition, their different fighting styles converge. Strikers face wrestlers. Boxers face grappling artists. Muay Thai and jujitsu practitioners square off. As the saying in the sport goes, “Styles make fights.”

Fighters are acutely aware of each other’s strengths and weaknesses. They know whether their opponent has heavy hands (like Conor McGregor’s famous left hand) or a penchant for a particular submission technique (like Ronda Rousey’s storied armbar). Because of these different styles, fighters are forced to think about how they are going to win. They can’t simply try harder. They must think about how they are going to use their strengths and exploit their opponent’s weaknesses to achieve victory.

When you watch these athletes train, particularly as they enter a training camp in preparation for a big fight, their coaches home in on one question: “How do you see the fight ending?” The fighter, having already rehearsed the scenario in her head throughout practice, knows the answer: “I win by knockout,” “I win by submission,” or “I win by armbar.” The “I win” part for MMA fighters is the outcome goal. The “by knockout,” “by submission,” or “by armbar” part peers into the process of achieving that goal. In training, MMA fighters rehearse how to control position, cut off angles, and physically wear out their opponent as different pieces of the process of accomplishing the outcome of winning. During the fight, pieces of the process they have rehearsed come to light as the fighters move, strike, and control position within the octagon. These fighters know it is not good enough to simply answer their coach’s question with “I win.” Their answer specifies how they will defeat their opponent.

As in MMA, defining the outcome goal in ultrarunning is only half the battle. The remaining part relies on setting process goals that will lead to success. Planning process goals is the actionable portion of your race-day goal-setting exercise. Unlike the outcome goal, which will be influenced by variables outside of your control, you have greater control over process goals come race day. They are the actions, thoughts, checkpoints, and supporting activities you do during the race to better ensure the outcome goal is achieved. In this way, process goals should dominate your thought process during the race. Thinking “I need to eat another gel to hit 200 calories an hour” at mile 30 of a 50-miler is going to do a lot more good than thinking “I need to finish in nine hours.” As with outcome goals, process goals need to be personalized to suit you as an individual, your capabilities, and most importantly your personality.

When I have my athletes move through the process of goal planning, highly individualized paths emerge to support their outcome goals. Many times, even athletes with very similar outcome goals have wildly different process goals. Your process is your own, but all well-set process goals meet the following criteria:

- The process goal must directly support the outcome goal.
- The runner must be able to control the process.

- The process goals must suit the runner's individual needs.

PROCESS GOALS MUST SUPPORT THE OUTCOME GOAL. As you move through the development of your race-day process goals, the most important question to answer is, "Do the process goals support the outcome goal?" Accomplishing each process goal should bring you one step closer to your outcome goal. Goals related to pace, effort level, attitude, nutritional planning, and gear selection are all valid process goals as long as they directly improve the likelihood of accomplishing the outcome goal. Process goals that do not directly affect the outcome are frivolous and only suck valuable energy from your race. A great litmus test for a properly constructed set of process goals is to run them by your crew. If a process goal is well constructed, it should be something that your crew can easily reinforce while you are in an aid station, as well as something that makes an impact down the road.

PROCESS GOALS MUST BE UNDER THE RUNNER'S CONTROL. For a process goal to be effective, you must be able to execute it during the course of race day. Process goals such as eating a certain number of calories, running at a particular pace, and having a great attitude are all under your control. Process goals that fall outside of your control, particularly if they are dependent on another person, are not properly set process goals. "Run with Analisa for the first half" might sound like an easy goal to accomplish, but you are not in control of Analisa's running. Avoiding the weather, staying in the top twenty, or beating a fellow runner are all similarly inappropriate; you as the runner are not in complete control.

HOW YOUR CREW CAN REINFORCE YOUR PROCESS GOALS

Outcome goal: Finish the American River 50.

Process goals: Eat 200 calories an hour.

List your process goals on a cue card and give it to your crew. Have them ask questions related to the specific process goals you need to go through, such as, "How many calories have you taken in? It's three hours into the race, so you should have consumed 600 calories by now."

PROCESS GOALS MUST SUIT THE NEEDS OF THE INDIVIDUAL. When setting process goals with my athletes, I am careful to help them tailor their goals to their individual needs, psychology, and personality. Many of the process goals you develop will include technical aspects (such as pace, calories per hour, and anything that can be quantified) and psychological facets of the event (such as "stay positive"). The emphasis on either technical or psychological aspects must be tailored to the individual, but I always encourage athletes to include at least one of each. If you run well when you track stats and numbers (such as calories and pace), you should have a larger proportion of those types of process goals. If you run well with more psychological cues (such as staying positive, having fun, interacting with the crew), these should constitute a larger proportion of your process goals. Similarly, when I am out at races encouraging athletes, I will provide the statistic-focused athlete with comments such as, "You ran that section right on your target time! Great job!" In contrast, an athlete who thrives more from psychological motivation will hear, "Stay positive! This next section is your strength, and your crew is right around the corner!"

Goal-Setting in Your First Ultra or First 100-Mile Event

The Leadville Trail 100 is notorious for having an extremely poor finish rate. Most years it hovers around 50 percent; some years, it's even lower. While the race is certainly difficult, there are no good reasons for this level of failure in the event. Some runners blame the altitude. Others say the climbing is just too much. Neither of these is true. As race founder Ken Chlouber would say, "These are crybaby excuses."

The real rationale for Leadville's poor finish rate revolves around the fact that there are no prerequisites to enter the race, as there are for many other 100-mile events. An athlete needs only to survive the lottery process, plunk down an entry fee, and show up in order to start the race. As a result, many athletes who enter the event are new to the sport or are using Leadville as their first 100-mile race. Many of the innocent rookies drop like flies. The effort required to complete 100 mountainous miles at high altitude torments runners as they attempt to locomote through the wee hours of the night and into the next day. The athletes struggle with cutoffs, calories, and a sense of what the hell they have gotten themselves into. Sadly, many who are struggling will refuse to continue. It's a tough race for sure, but not as tough as one would expect from a 50 percent finish rate.

Far more than 50 percent of the athletes entering the Leadville Trail 100 have the necessary fitness to complete the race. The athletes who don't finish rarely lack the physical ability, but rather a sense of purpose. Purpose is what will drive them in and out of the frigid Outward Bound aid station at mile 75 when a warm car ride back into town is oh so tempting. The rookie mistake is not a lack of training, it is a failure to understand the special feeling that comes from finishing one's first 100-mile race. They come to the race without that singular purpose, and it costs them.

It has taken me until this point in the book to insert a personal story about training and preparing for ultramarathons. All the earlier chapters are filled with anecdotes from my athletes, and this is on purpose in order to avoid introducing my own N-of-1 bias.

My first 100-mile event was the Leadville Trail 100 in 2008. An athlete's first ultra and first 100-mile race are special moments. These two accomplishments represent unique achievements that are unparalleled in the runner's athletic career. Because one's "first" is such a special moment, the goal-setting process for these events should revolve around getting to the finish line at all costs. Time, place, and performance in these circumstances are all secondary considerations.

I will always remember the year I first finished the Leadville Trail 100, because I proposed to my wife at the finish line. I chose to reinforce the "finish at all costs" mentality by carrying a ring in the pocket of my shorts for the entirety of the 2008 Leadville Trail 100. For the finish line proposal, I had no backup plan. I either finished the race and got on one knee, or . . . well, I didn't know what the hell I was going to do. For 100 miles, the ring was there, with the center stone occasionally pressing into my thigh. It was a constant reminder of how important the race was and how special the finish would be. That was my outcome goal—finish the race and have the chance to propose to my future wife. There were no B, C, or D goals, just one goal and one focus.

My process goals were similarly suited: run conservatively, take time in the aid stations, enjoy the company of my pacers, and take additional warm gear. All were aimed at finishing in any time under the thirty-hour cutoff. While the race was hard, the ending of the story was a happy one. Sure, I was well prepared and fit, and I had a fantastic support team (including this book's coauthor, Jim Rutberg). But the crucial aspect of preparation was setting the all-important outcome goal and its supporting process goals. Achieving that goal meant the world to me.

The Dreaded DNF

Ultramarathons are hard. Most experienced ultrarunners have at least one “Did Not Finish,” or DNF, on their record. Sure, you may come across athletes without a DNF, but these individuals are at the edges of the bell curve. In my coaching career, I have had my fair share of athletes who have DNFed in an ultra. I have had several DNFs myself. Some could have been prevented, but others were unavoidable. While it’s always heartbreaking to see an athlete underperform, I have never, not once, been upset with, ashamed by, mad at, or embarrassed by an athlete who takes a DNF.

There are three reasons for this:

- If a DNF results from some random act like rolling an ankle or getting off course, that’s part of the game. No sense in agonizing over it. Injuries and mishaps can happen on any training day and on any training mile throughout the year. It’s just dumb luck if it happens on race day.
- If a DNF results from a lack of preparation, either physical or mental, that’s on me, provided the athlete did the work in training. It’s my job to make sure the athlete gets to the starting line prepared for the event. Similarly, my hope in writing this book is to give you enough knowledge and tools so you will be well prepared when you reach the starting line for your event.
- Some athletes choose to race at the edges of their capabilities. That is their goal. For these athletes, a DNF may be the right personal choice once their aggressive goal slips away, even when completing the race is still a possibility. Reaching the finish line might require limping through the remaining miles or a three-hour nap at the aid station; they could still finish, yet they choose not to continue. These athletes have chased the goal they set and should have no shame in that DNF. If you are clear about your goals, then a DNF as a potential ramification of aggressive goal setting is just as gutsy as struggling to the end. As long as you are crystal clear on your goals, find the right balance of challenge and risk, and take the time to craft goals in a meaningful way, then you will always make the right decision if faced with the possibility of a DNF.

Creating an Effort-Based Race-Day Strategy

Pacing can make or break your race day. Wrist-top GPS technology can provide accurate information on pace, altitude, cadence, and heart rate to help guide the process. These data cues can be used to better prepare for race day, but reliance on this information can also ruin your plans if you apply it incorrectly.

Pacing data from athletes competing in the Western States 100 reveal that athletes who perform better and place higher generally have less speed variability overall and slow down less as the race progresses (Figure 16.6; Hoffman 2014). The easy conclusion to make is that these athletes paced the race better and thus slow down less and finish the race faster. However, could it be that the athletes who slowed down less were just more prepared? That question has yet to be answered.

In any case, there are three different strategies to properly gauge your effort and speed during a race: pace, heart rate, and perceived exertion. You can choose to utilize one or a combination that suits your needs and situation.

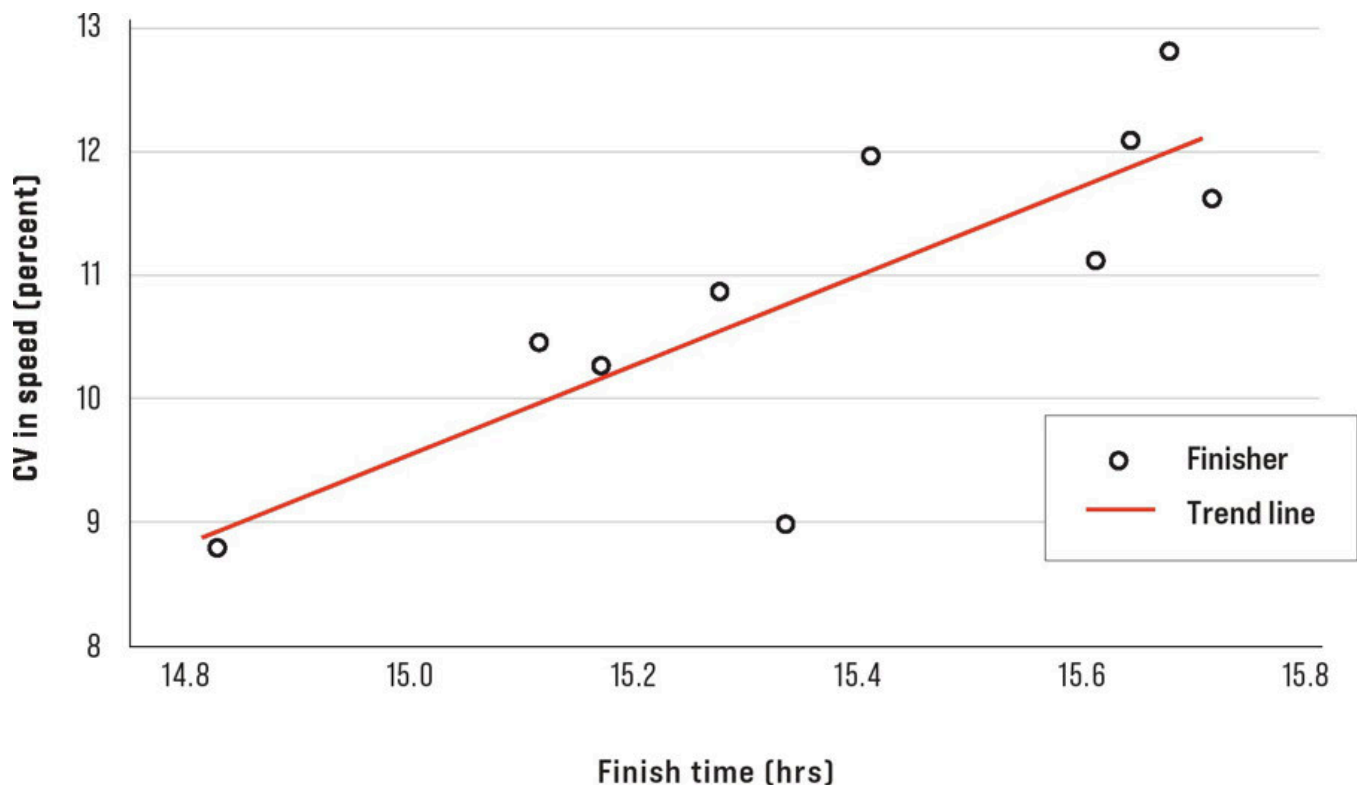


FIGURE 16.6 Relationship between coefficient of variation (CV) in speed and finish time for the ten fastest finishers of the Western States 100. The fastest finishers had the lowest variation in speed. Source: Hoffman 2014.

OPTION 1: Pace

The standard marathon race-day plan invariably revolves around hitting some sort of pace for specific segments of the race. Even before the advent of GPS watches, digital timing clocks lined the routes of the major marathon courses, sometimes appearing as often as every mile. As GPS watches evolved, runners started monitoring pace in real time. For athletes looking to set a particular pace or time standard for a marathon, this is a good solution. You have all the data you need at regular intervals to pace the race out properly. In many major marathons, you can even get customizable pacing bracelets, complete with splits for each and every mile.

As I've said already, ultramarathons are not simply long marathons. Thus, although pace can be a useful component of your race-day strategy, there are limitations that you will have to account for to determine if it is a useful tool for you and your goals.

Pace is a good tool on race day in the following scenarios:

- Race terrain is flat and level, without any technical elements for 50K and 50-mile distances. In this situation you can choose to base your race-day strategy entirely on pace. If you are keenly aware of your TempoRun, SteadyStateRun, and EnduranceRun paces, your race-day paces can mirror these when appropriate. Similarly, if there is an element of flat, level terrain somewhere within a trail race, you can incorporate pace in those sections to gauge your race-day intensity.

- You are able to train on the racecourse. In this situation, if you are aware of your pace on certain segments and during certain workouts, you can use that information to calibrate your race-day pace. For example, if you consistently do SteadyStateRun intervals up a climb in training and that climb is used during the race, you can check your GPS watch at various points during the climb to make sure you are at the right pace. For 50K and 50-mile races many athletes can climb segments at their SteadyStateRun or TempoRun intensities. Generally speaking, if the total number of climbing segments in the race matches the training interval lengths described in Chapter 9 (see Table 9.1), you can reproduce those same intensities during the race if you have trained properly for them.

Pace is a poor tool on race day in the following scenarios:

- The terrain is different than your home training ground. The terrain will slow you down if it is more technical than you're accustomed to, and the opposite is true if the race-day terrain is more benign. This is true at all race intensities and at all grades. Bottom line: if your race-day terrain is much more or less difficult than your home training ground, pace will be a poor tool for you come race day.
- The altitude is different than your home training ground. When you are at higher altitudes, your pace will be negatively affected. The opposite is true if you are racing at lower altitudes. These effects are highly individual, making it difficult or unwise to attempt to adjust pace based on a mathematical formula.
- For 100-kilometer and 100-mile events. Typically these events are so long that you will be running much slower and easier than your day-to-day training paces and intensities.

OPTION 2: Heart Rate

As discussed in Chapter 6, heart rate is generally a poor way for an ultrarunner to determine intensity. Factors such as sleep, temperature, hydration, fatigue, and time of day may affect heart rate at any particular point in time. Furthermore, because I am an advocate of training based on RPE, introducing heart rate on race day would violate the tried-and-true rule of "try nothing new on race day." Nevertheless, there is one limited scenario in which it is appropriate to use heart rate to gauge your race-day intensity.

Heart rate is a good tool on race day in the following scenario:

- When an athlete is consistently overenthusiastic at the start of races. In this case, I recommend using a heart rate monitor as a governor for the first 25 percent of the race. The simple procedure I use is to find the athlete's normal SteadyStateRun heart rate range during training (still using RPE to determine the intensity during the actual workouts) and then have them use that value as an upper limit that is not to be exceeded during the first 25 percent of the race. For example, if during training an athlete's heart rate range is normally 160 to 165 beats per minute for SteadyStateRun intervals, I will set 165 as the heart rate ceiling that the athlete cannot exceed for the first 25 percent of the race. Effectively this holds the athlete at the lower end of SteadyState intensity or higher end of EnduranceRun, because race-day heart rates are generally higher for the first 25 percent of the race due to freshness and adrenaline.

Heart rate is a poor tool on race day in the following scenarios:

- Just about everything else. Your race-day heart rate is going to be affected by a multitude of factors. Early in the race, adrenaline will artificially elevate heart rate. As fatigue sets in, your heart rate will be depressed. All in all, heart rate is a poor choice to use come race day.

OPTION 3: RPE

Racing should be a reflection of your training. Similarly, pacing during the race should revolve around how you pace your day-to-day efforts. With my athletes this means using RPE. Your internal RPE offers a calibration point that accurately identifies your intensity and is independent from factors that affect pace and heart rate. But the main reason my athletes use RPE on race day is because it is what they have used in training. Day in and day out, they are calibrating their efforts using a simple 1 to 10 scale. Race day should be no different. (For a detailed look at RPE, refer to Chapter 6.) The sidebar “Calibrating RPE for Race Day” illustrates how an athlete was able to take his efforts during day-to-day training and translate them into proper pacing on race day.

CALIBRATING RPE FOR RACE DAY

IF YOU HAVE TAKEN THE TIME to properly train and develop your fitness, you can race as you have trained. This means that during the race you should be able to recreate the time-at-intensities you accomplished in training. For example, if you were consistently doing four by ten-minute efforts at TempoRun intensity during training, you can accumulate forty to sixty minutes of the same intensity during the race. The same is true for EnduranceRun and SteadyStateRun intensities. Based on the attributes of the race you are preparing for, you can map out a plan of what RPE to target for different parts of the race.

The first figure (A) is a typical TempoRun that an athlete did in preparation for the 2014 North Face Endurance Challenge. It shows that he can average a Normalized Graded Pace (NGP) of 5:30–6:00 min/mi for three by eleven-to-twelve-minute intervals. The intervals for this workout would be at an RPE of 8 to 9.

The second figure (B) is a GPS file from the same athlete’s actual race. The first four major climbs of the race are highlighted with the NGPs the athlete was able to run. In this situation, he accurately calibrated his race-day effort from his training efforts because the climb segments were similar in length to what he was doing in training. The race data validate this strategy, as the NGP from the first three climbs is similar to his TempoRun NGP during training. The NGP from the fourth climb would be indicative of a SteadyStateRun effort because it is closer to sixty minutes in length.

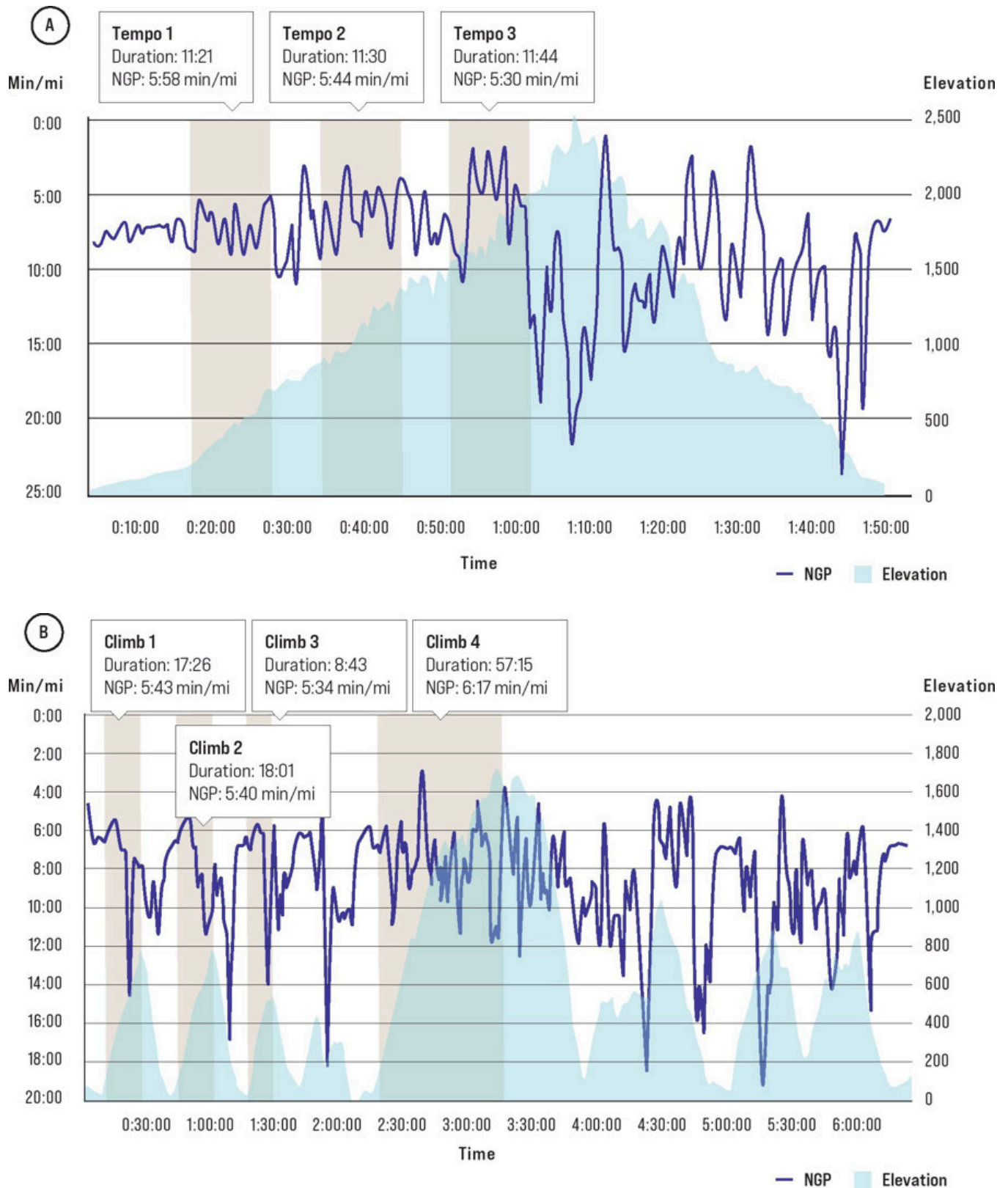


FIGURE 16.7 Comparison of Tempo intervals done in (A) in training to (B) the climbs in a race.

Why Staying in the Moment Works

“Focus on the mile you are in.” “Just take it one aid station at a time.” “Be present.” If you have been at an aid station during an ultra, you have almost certainly heard these strategies, and others, uttered by pacers and athletes as a means of psychologically coping during the race. They’ve been so overused that they are trite by now. At their roots, they are all trying to break up a long effort into more manageable shorter ones, otherwise known as “compartmentalizing the effort.”

Few think about why this strategy is actually effective; they’ve just regurgitated it based off of an experience, anecdote, or tall tale. But it turns out there’s a scientific rationale to compartmentalizing the effort. Additionally, this practice is well suited for ultrarunning because of the sport’s complexity and uncertainty.

HOW WE PACE WORKOUTS

Ultrarunners have always adopted a wide range of pacing strategies. While I advocate for a perceived exertion–based strategy for trail and ultrarunners, I realize my opinion is not the only game in town. Some athletes choose to use heart rate, others utilize pace, and some runners are starting to use foot-based power meters. When I was in college, our coach would place orange cones every 50 m along the track and then program a beeper to broadcast over the loudspeakers at distinct intervals corresponding to the paces we were trying to achieve. All of these strategies have their merits: they provide feedback to the athlete, and the athlete can either adjust their effort or maintain it accordingly. However, when uncertainties enter the equation—as they do in ultrarunning—the athlete has to apply a more nuanced understanding of pacing.

A 2018 study out of Australia looked at athletes performing a 30K cycling time trial with and without cues about the distance remaining (Wingfield et al. 2018). Athletes who knew how far they had gone in the time trial were able to achieve higher power outputs (particularly in the last two minutes) despite no changes in heart rate or RPE (see Figure 16.8).

This makes a lot of sense. When you know how much farther you have to go, you can take an internal risk assessment to gauge how hard you can push (see Figure 16.9). This phenomenon is what is known as “perceived exertion end point interaction.” It simply states that if given a task, let’s say a workout or a race, athletes will pace themselves according to two facets: (1) how they feel at the moment and (2) how far they have to go. You constantly integrate those two points as you progress through the task and calibrate your effort so you are spent (or at least perceive that you are spent) at the very end. You are literally drawing and redrawing an internal line between where you feel right now and where you project you will feel at the end point.

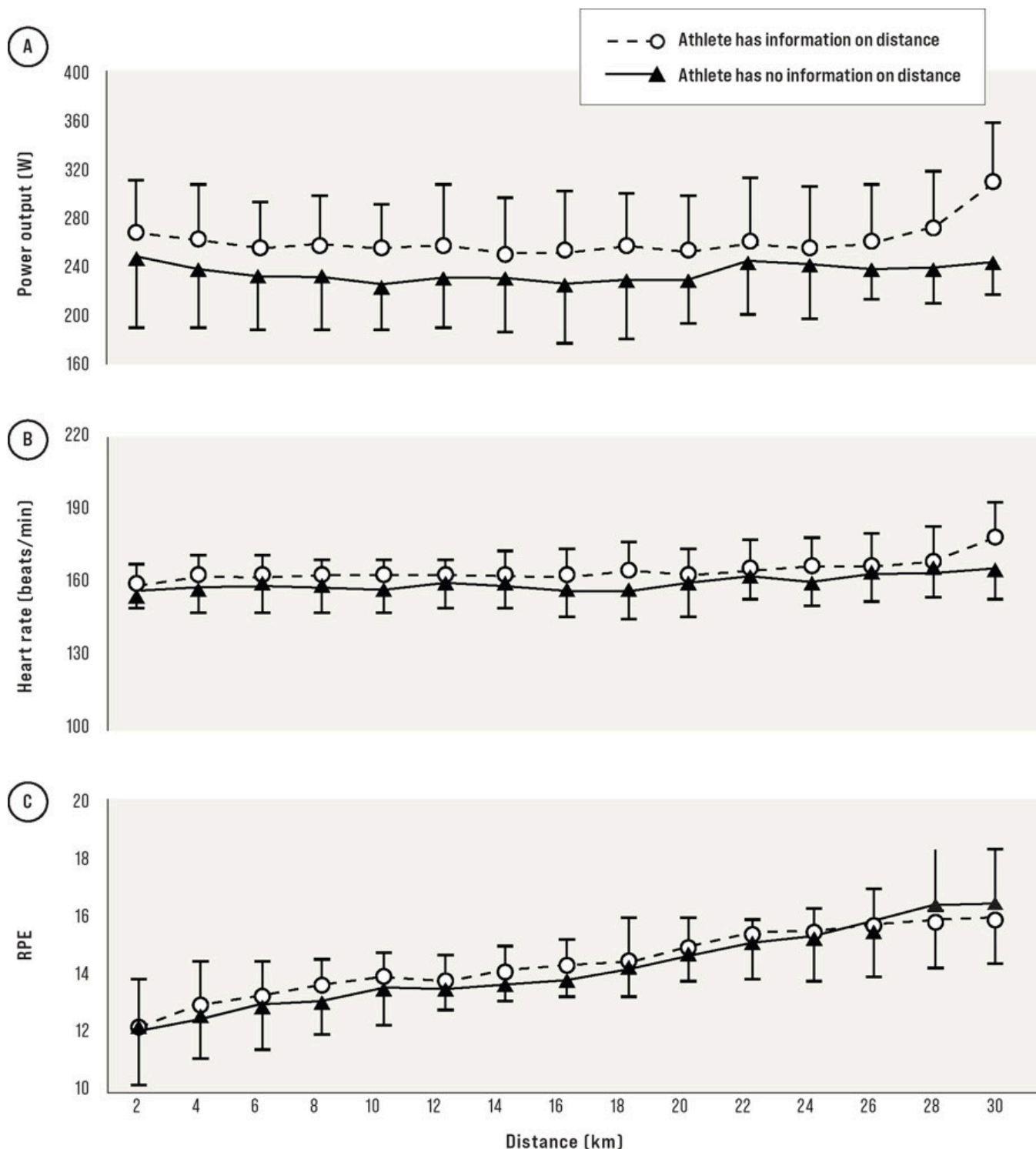


FIGURE 16.8 A 30K cycling time trial done with information on distance and without. Source: Wingfield 2018.

Interestingly, this rationale has also been used to explain why novice athletes are so poor at pacing, because they don't know the end point well enough. It can explain how "tougher" athletes can perform better; they are willing to risk a steeper slope and higher end point (see Figure 16.10). And it also explains why youth athletes have a tendency to start out too hard, as they have poor internal measurements of exertion. Flavors of this phenomenon have also made their way into famous models of performance regulation, such as Tim Noakes's central governor theory (Noakes, Peltonen, and Rusko 2001) and Ross Tucker's Anticipatory Regulation Model (Tucker 2009). I also feel there is a translational point in ultrarunning.

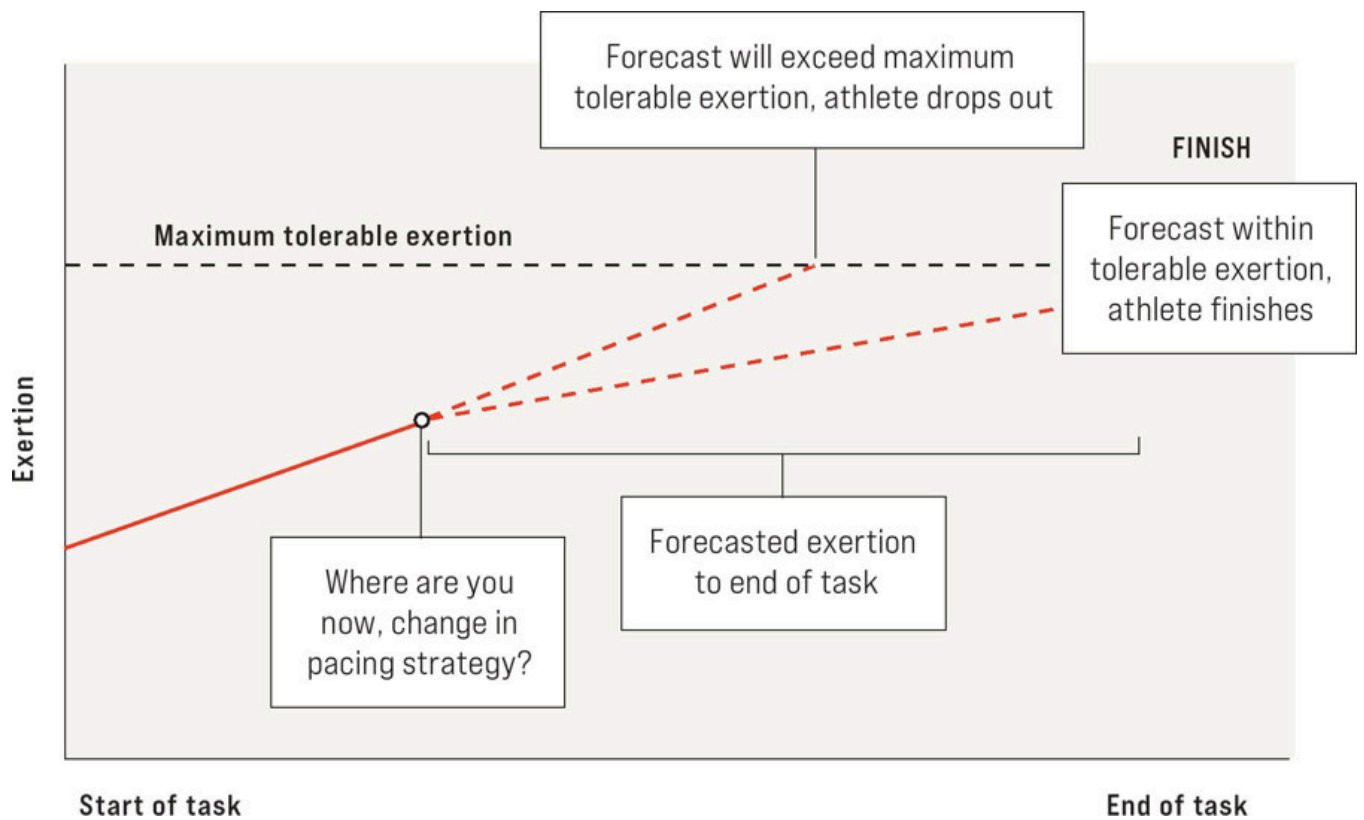


FIGURE 16.9 Perceived exertion end point interaction.

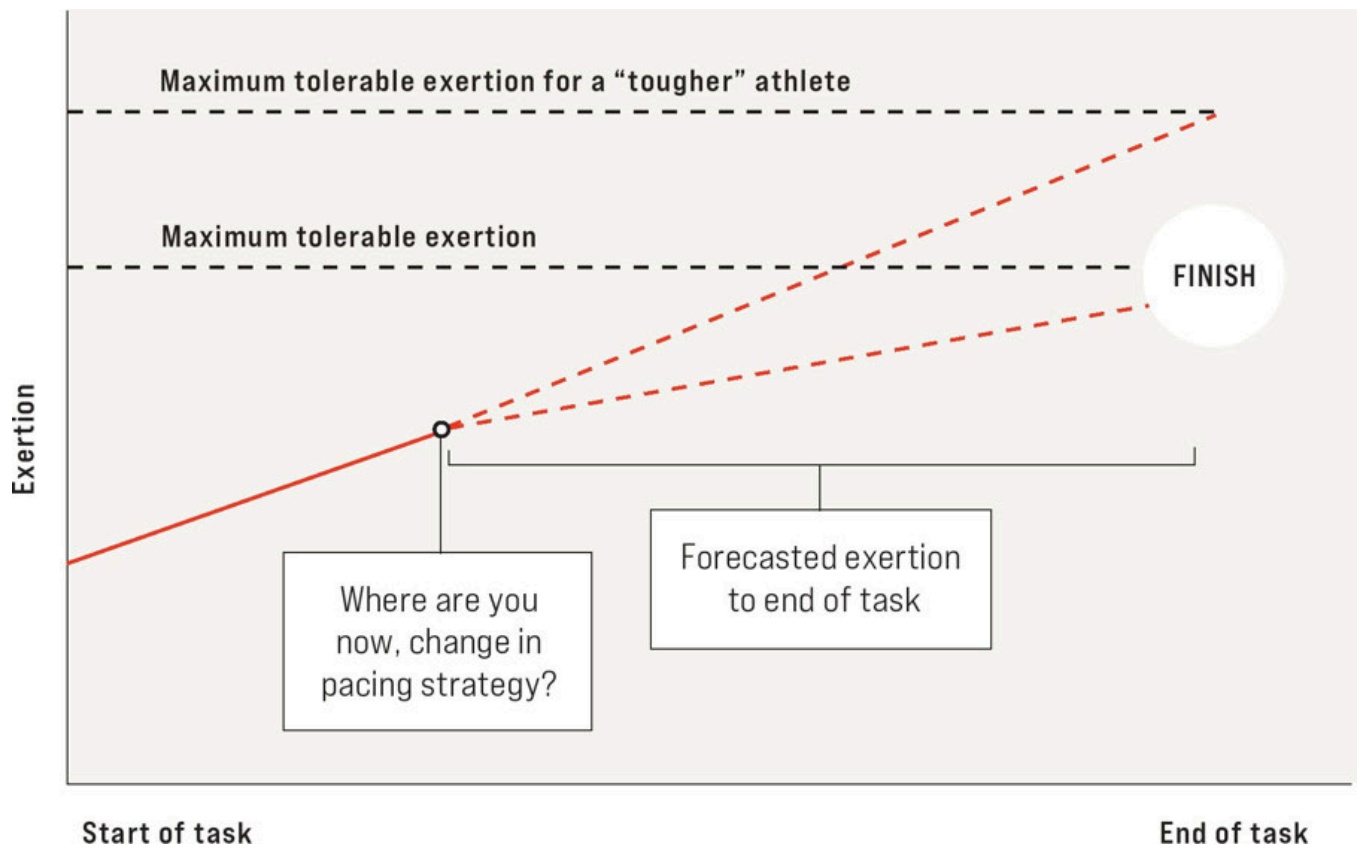


FIGURE 16.10 How a "tougher" athlete fares better.

FORECASTING DNF DECISIONS

If you have been keeping track during this chapter, I've used the dreadful DNF rates at the Leadville Trail 100 as an example of poor goal setting. There's also much to be learned from observing how and where athletes make the choice to drop out. Every year I make it a point to spend several hours at the Outward Bound aid station at the Leadville Trail 100. Located at mile 75 of the race, it is a DNF decision point for many. Runners come into the aid station in all different states. Some look great, others awful. Despite their varying outward-looking conditions, there is no real correlation between the athletes' appearances and who drops out or continues on to the finish. There is, however, an internal calculation made by each and every person as they enter that aid station. Whether they realize it or not, they look at how they feel at the moment and extrapolate that to how they anticipate feeling while running the remaining twenty-five miles back to the town of Leadville—which include the infamous Powerline climb. According to the perceived exertion end point interaction concept, if the slope or the end point of that imaginary line exceeds what they perceive is tolerable to them, they get their wristband cut, head into their car, and take the short ride back to town. The athletes that forge on have imaginary lines that are within their preconceived limitations, whether through a higher exertion tolerance, a more accurate forecast, or a lower degree of instantaneous suffering. However, the potential for error in both of these strategies remains the same: you are forecasting for a long period of time with many unknowns.

IN THE MOMENT

The take-home message for ultrarunners is that there's some method to the madness of "being in the moment." Forecasting and predicting in ultrarunning can be a fool's errand, although we're naturally inclined to do it. We're even trained to incorporate perceived exertion end point interactions in workouts and hard runs. However, unlike in a workout or marathon, your ability to predict how you will feel hours down an ultramarathon trail is clouded by the sheer duration and your impaired ability to comprehend (because let's get real: are you smarter or dumber after eighty miles?). So, when you get to your own Outward Bound aid station and all seems hopeless, forget the forecasting. Stay in the moment, focus on your effort at the present time, and draw upon some of the mental skills that we covered in Chapter 15. The finish line will eventually show up—just give it time.

Sleep Considerations for Longer Events (200 miles, etc.)

Ultramarathons lasting more than 48 hours require some special planning, particularly in regard to sleep. This is an area where the research is pretty thin, and we are doing a lot of educated guesswork in terms of when, where, and how much to sleep during events. Rather than just leave it at "it depends," let's go over the basics of what we know about when, where, and why the sleep monster might jump on your back.

THE WITCHING HOUR

Ultrarunning folklore is riddled with tales of hallucinations, out-of-body experiences, paranormal activities, and even an alien sighting or two. Given enough time and sleep deprivation, ghouls and goblins, devils and demons, witches and wildlife will inevitably appear, haunting even the heartiest of souls. I've been there. Deep into the Tor des Géants I swore some random Italian at an aid station was one of my lifelong friends. The bewildered look on